

Hull Tactical Market Prediction

Tactical Asset Allocation with Machine Learning

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The Kaggle Challenge

- **Goal:** predict daily S&P 500 excess returns and output a weight $w \in [0, 2]$
- **Metric:** adjusted Sharpe ratio, penalizes both excess volatility and return underperformance
- **Data:** `train.csv` only, $\sim 9k$ anonymized trading days (macro, sentiment, pricing, volatility features)

The adjusted Sharpe metric is what shapes the whole strategy: staying close to the index ($w = 1$) is often better than a confident wrong prediction.

Adjusted Sharpe Ratio

$$\text{adjusted_sharpe} = \frac{\text{Sharpe}}{\text{vol_penalty} \times \text{return_penalty}}$$

$$\text{vol_penalty} = 1 + \max\left(0, \frac{\sigma_{\text{strat}}}{\sigma_{\text{mkt}}} - 1.2\right)$$

$$\text{return_penalty} = 1 + \frac{\text{gap}^2}{100}$$

$$\text{gap} = \max(0, (\mu_{\text{mkt}} - \mu_{\text{strat}}) \times 100 \times 252)$$

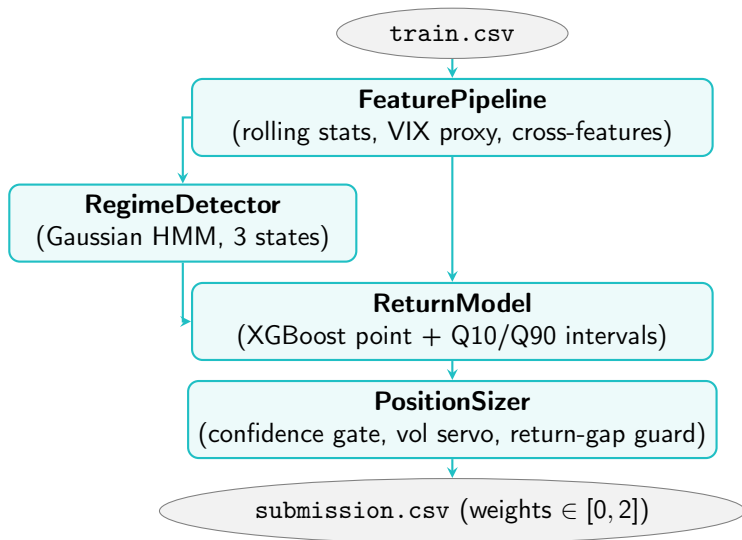
The **quadratic** return penalty is what makes sustained underperformance really hurt, falling 5% behind the market annualized multiplies the denominator by ~ 1.25 . This is why I default to $w = 1$ unless the signal is genuinely strong.

- 8 990 trading days: forward returns, risk-free rate, and anonymized features
 - I1–I5 (interest rates), E1–E12 (economic), S1–S5 (sentiment), P1–P11 (pricing), V1–V13 (volatility)
- Split chronologically into three non-overlapping blocks:

Tuning block	Purge gap	Holdout
7 290 rows	200 rows	1 500 rows

The 200-row purge gap sits between tuning and holdout to prevent autocorrelation leakage at the boundary.

System Overview



Rolling Statistics

- Volatility: $\sigma_{5,20,60,252}$
- Momentum: $r_{5,20,60,252}$
- Sharpe ratios: $SR_{20,60}$
- Drawdown: DD_{60} , $MaxDD_{252}$

VIX Features

- VIX Proxy: mean of V1–V13
- Percentile rank (train-frozen)
- Term-slope: $V13 - V1$
- VIX delta (history-buffered)

~107 engineered features after availability filtering.

Cross-Features

- Yield-curve proxy: $I_{last} - I_{first}$
- Macro-sentiment divergence
- Price-momentum agreement
- Autocorrelation: $\rho_{5,20}$

Anti-Leakage

- Median imputation from train only
- 400-row history buffer for rolling windows
- Lagged targets only: `.shift(1)`

The V1–V13 columns are anonymized, but I treat their cross-sectional mean as a VIX-like fear gauge:

$$\text{VIX Proxy}_t = \frac{1}{13} \sum_{i=1}^{13} V_{i,t}$$

From this single signal I derive four features fed to both the model and the regime detector:

- 1 **VIX proxy**, the raw level
- 2 **VIX percentile rank**, where it sits in the training distribution (frozen)
- 3 **VIX term-slope**, $V13 - V1$, a rough volatility term structure
- 4 **VIX delta**, day-over-day change, momentum of fear

I fit a **Gaussian HMM** with 3 hidden states and diagonal covariance on four features: `vix_proxy_pct`, `roll_vol_20`, `momentum_20`, `autocorr_20`.

Each row gets:

- `regime_state`, Viterbi-decoded hidden state
- `regime_prob_{0,1,2}`, posterior probabilities
- `regime_stability`, confidence in the current assignment
- `regime_label`, semantic label (`bull_low_vol`, `bear_high_vol`, `transition`) assigned by comparing training-set mean returns and volatility across states

Return Prediction Model (XGBoost)

I train three XGBoost models simultaneously on `market_forward_excess_returns`:

Model	Objective	Purpose
Point model	reg:squarederror	Direction & magnitude
Lower quantile (Q10)	reg:quantileerror, $\alpha = 0.1$	Lower bound
Upper quantile (Q90)	reg:quantileerror, $\alpha = 0.9$	Upper bound

Confidence signal: how large the point prediction is relative to the interval width,

$$\text{raw_conf} = \frac{|\text{point_pred}|}{\max(\text{Q90} - \text{Q10}, \varepsilon)} \quad \longrightarrow \quad \text{confidence} = \text{clip}\left(\frac{\text{raw_conf} - p_{10}}{p_{90} - p_{10}}, 0, 1\right)$$

p_{10}, p_{90} are computed on training data and frozen, mapping confidence into $[0, 1]$.

XGBoost parameters

- `max_depth = 4`
- `learning_rate = 0.05`
- `subsample = 0.8`
- `colsample_bytree = 0.8`

Tree counts

- Point model: **400 trees**
- Quantile bounds: **150 trees** each, shorter ensembles reduce overfitting to noisy tails

Purged walk-forward CV: expanding training windows with a configurable purge gap before each validation fold. No information crosses the boundary through autocorrelated targets.

Confidence-Gated Position Sizing

Weight rule:

$$w = 1.0 \quad \text{if conf} < \text{threshold}$$
$$w = 1.0 + \text{gain} \cdot \hat{r} \cdot \text{vol_scale} \cdot \text{reg_mult} \quad \text{otherwise}$$

Volatility scaling (Moreira & Muir, 2017):

$$\text{vol_scale} = \text{clip}\left(\frac{\sigma_{\text{target}}}{\sigma_t}, 0.5, 2.0\right)$$

When confidence is low I stay at $w = 1$; otherwise I scale risk inversely to volatility and reduce exposure in adverse regimes.

Regime multipliers

Regime	Mult.
Bull, low vol	1.30
Normal	1.00
Transition	0.70
Bear, high vol	0.50

Two guards run sequentially after the base weight is computed:

- **Volatility servo:** if rolling strategy vol / market vol exceeds the soft cap (1.149) or hard cap (1.20), I scale the active deviation back to 55% or 20% respectively, keeping the vol penalty from triggering.
- **Return-gap guard:** if the strategy has fallen behind the market, I force $w \geq 1.0$ to stop the gap from widening and avoid the quadratic return penalty compounding.

Both guards are defensive by design, they reduce downside without capping upside on days where the signal is strong and volatility is in range.

Walk-Forward Backtesting

Each CV fold re-fits the **entire pipeline** from scratch: features, regime detector, and return model. No state is shared between folds. I cache the model predictions per fold so Optuna only needs to re-run the lightweight weight-generation step.

Fold	Validation Range	Adj. Sharpe	Mean Weight	Dominant Regime
1	3 790 – 4 372	+0.9985	0.914	Transition
2	4 373 – 4 955	-0.2741	1.136	Transition
3	4 956 – 5 538	+0.8578	0.981	Bear (High Vol)
4	5 539 – 6 121	+1.5707	1.012	Transition
5	6 122 – 6 704	+0.9661	0.950	Bear (High Vol)
6	6 705 – 7 287	+0.9965	0.930	Transition

I fix the modeling pipeline first and tune only the four strategy parameters (confidence threshold, gain, target volatility, soft vol cap) using Optuna with 40 trials.

- Untuned baseline mean CV adjusted Sharpe: **0.7246**
- After tuning: **0.8526**

Caching the fold predictions is important here: fitting the feature pipeline and XGBoost models takes most of the compute, so running 40 Optuna trials on top of cached predictions is fast and fully reproducible.

Holdout Results

Configuration	Adj. Sharpe	Ann. Return	Vol Ratio	Active Days
This model (Tuned)	0.8605	+20.2%	1.020	29.5%
Baseline ($w = 1.0$)	0.7151	+16.8%	1.000	—
Delta	+0.1454	+3.4 pp		

- Only 29.5% of days are active, the confidence threshold is deliberately high so most days I just hold the index
- Tuned parameters: confidence threshold = 0.364, gain = 179.3, target vol = 0.298, soft vol cap = 1.149

Leakage Prevention

Risk	Mitigation
Feature leakage from future Rolling boundary leakage	<code>fit_transform</code> on training only; apply frozen stats 400-row history buffer prepended to validation/holdout blocks
Temporal lookahead in CV	Purged walk-forward CV; separate 200-day purge before final holdout
Target leakage	Lagged targets only (<code>.shift(1)</code>); target excluded from features
Tuning lookahead XGBoost non-determinism	Strategy parameters tuned on inner temporal holdout Enforce <code>nthread=1</code> , <code>tree_method='hist'</code> , <code>seed=42</code>

Any one of these omissions would inflate the backtest. The purge gap and history buffer are the two that are easiest to miss.

Reproducibility:

- All random seeds set to 42 (XGBoost, sklearn, HMM)
- `nthread=1` and `tree_method='hist'` for cross-platform determinism

Why only 29.5% active days?

Most days the Q10–Q90 interval is wide relative to the point estimate, so raw confidence is low and I stay at $w = 1$. Active days tend to cluster around trend-consistent predictions in low-volatility bull periods, exactly when the model's signal-to-noise is highest and the volatility penalty is least likely to trigger.

- The quadratic return penalty in the adjusted Sharpe is the main design constraint, it forces a conservative default.
- Confidence-gating on the quantile interval width is what keeps the strategy from overtrading on noisy predictions.
- Volatility scaling (Moreira & Muir, 2017) and regime multipliers modulate the active bet size without changing the direction.
- Purged walk-forward CV with cached predictions makes the tuning stage both fast and leak-free.

- Moreira, A. & Muir, T. (2017). *Volatility-Managed Portfolios*. Journal of Finance, 72(4), 1611–1644.
- Kaggle Competition:
<https://www.kaggle.com/competitions/hull-tactical-market-prediction>
- Chen, T. & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD 2016.